

# BV-CHP

BV-CHP is a combination of cancer drug. It is used to treat a rarer type of non-Hodgkin lymphoma called systemic anaplastic large cell lymphoma.

## On this page

- [What is BV-CHP?](#)
- [How BV-CHP is given](#)
- [About side effects](#)
- [Side effects while treatment is being given](#)
- [Common side effects](#)
- [Less common side effects](#)
- [Other information](#)
- [How we can help](#)

## What is BV-CHP?

BV-CHP is a combination of cancer drugs which is also sometimes known as B-CHP.

It is used to treat a rarer type of [non-Hodgkin lymphoma](#) called systemic anaplastic large cell lymphoma. It may sometimes be used to treat other cancers.

BV-CHP stands for the initials of the drugs used:

- BV – [brentuximab vedotin \(Adcetris<sup>®</sup>\)](#).
- C – [cyclophosphamide](#)
- H – [doxorubicin \(hydroxydaunomycin\)](#).
- P – prednisolone (a [steroid](#)).

Brentuximab vedotin is a [targeted therapy drug](#) called a monoclonal antibody. It delivers a chemotherapy drug directly to cancer cells.

It is best to read this information about this treatment with our general information about [chemotherapy](#) and the [type of cancer](#) you have.

Your doctor will talk to you about this treatment and its possible side effects before you agree ([consent](#)) to have treatment.

## More information about this treatment

This information is correct at time of publishing. But sometimes the types of cancer this treatment is used for, or treatment side effects, may change between revision dates.

You can talk to your cancer team if you want more detailed information about this treatment. Or visit the [electronic Medicines Compendium \(eMC\)](#) website, which has patient information leaflets (PIL) for individual drugs.

## How BV-CHP is given

You will usually be given BV-CHP in the chemotherapy day unit or in a clinic as an outpatient. Or you may have it during a stay in hospital.

During a course of treatment, you will see a:

- cancer doctor
- chemotherapy nurse or a specialist nurse
- specialist pharmacist.

This is who we mean when we mention doctor, nurse or pharmacist in this information.

Before or on the day of each treatment, a nurse or person trained to take blood (phlebotomist) will take a blood sample from you. This is to check that it is safe for you to have chemotherapy.

You will speak to a doctor, nurse or pharmacist before you have treatment. They will talk to you about your blood results and ask how you have been feeling. If your blood results are okay, the pharmacy team will prepare your cancer drugs.

Your nurse usually gives you anti-sickness (anti-emetic) drugs before the cancer drugs. You may have your treatment through:

- [cannula](#) – a short, thin tube the nurse puts into a vein in your arm or hand
- a [central line](#) – a fine tube that goes under the skin of your chest and into a vein close by
- a [PICC line](#) – a fine tube that is put into a vein in your arm and goes up into a vein in your chest
- an [implantable port](#) (portacath) – a disc that is put under the skin on your chest or arm and goes into a vein in your chest.

## Your course of treatment

You usually have a course of 6 cycles of treatment over a few months. Each cycle of BV-CHP takes 21 days (3 weeks). Your cancer team will talk to you about how the treatment will be given. They may give you a copy of a treatment plan to take home.

On day 1, the nurse will give you:

- prednisolone tablets
- cyclophosphamide, as a slow injection or a short infusion (drip) into a vein
- doxorubicin, as a slow injection into a vein or short infusion (drip) into a vein
- brentuximab vedotin, as an infusion over 30 minutes.

On days 1 to 5, you take prednisolone tablets.

You then have a rest period with no treatment for 16 days. This completes your first cycle of BV-CHP. At the end of the rest period, you start your second cycle of BV-CHP.

## About side effects

We explain the most common side effects of this treatment here. We also include some that are less common.

You may get some of the side effects we mention, but you are unlikely to get all of them. And you may have some side effects, including rarer ones, that we have not listed here.

Always tell your doctor, nurse or pharmacist about any side effects you have. They can give you:

- drugs to help control some side effects
- advice about managing side effects.

It is important to take any drugs exactly as explained. This means they will be more likely to work for you.

## Serious and life-threatening side effects

Some cancer treatments can cause serious side effects. Sometimes, these may be life-threatening. Your doctor, nurse or pharmacist can explain the risk of these side effects to you.

## Contact the hospital

Your doctor, nurse or pharmacist will give you 24-hour contact numbers for the hospital. If you feel unwell or need advice, you can call at any time of the day or night. Save these numbers in your phone or keep them somewhere safe.

## Side effects while treatment is being given

Some people may have side effects while they are having a cancer drug or shortly after they have it:

### Allergic reaction

Some people have an allergic reaction while having this treatment. Signs of a reaction can include:

- feeling hot or flushed
- shivering
- itching
- a skin rash
- feeling dizzy or sick
- a headache
- feeling breathless or wheezy

- swelling of your face or mouth
- pain in your back, tummy or chest.

Your nurse will check you for signs of a reaction during your treatment. If you feel unwell or have any of these signs, tell them straight away. If you do have a reaction, they can treat it quickly.

Sometimes a reaction happens a few hours after treatment. If you develop any of these signs or feel unwell after you get home, contact the hospital straight away on the 24-hour number.

**Always call 999 if swelling happens suddenly or you are struggling to breathe.**

## Pain along the vein

This treatment can cause pain:

- at the place where the drip (infusion) is given
- along the vein.

If you feel pain, tell your nurse straight away. They can check the site. They may give the drug more slowly or flush it through with more fluid to reduce pain.

## The drug leaks outside the vein

Sometimes cancer drugs that are given into a vein may leak outside the vein. If this happens, some drugs can damage the tissue near the vein. This is called extravasation. Extravasation is not common, but it is important that it is dealt with quickly. If you have any of the following symptoms around the vein during or after your treatment, tell your nurse straight away:

- stinging
- pain
- swelling
- if you have white skin, the area may become red
- if you have black or brown skin, the area may become darker.

## Common side effects

### Risk of infection

This treatment can reduce the number of white blood cells in your blood. These cells fight infection. If your white blood cell count is low, you may be more likely to get an [infection](#). A low white blood cell count is called neutropenia.

An infection can be very serious when the number of white blood cells is low. Your doctor may give you antibiotics and other drugs to take during this treatment. These may help reduce your risk of getting an infection.

But it is still important to get any infection treated as soon as possible. If you have any of the following symptoms, contact the hospital straight away on the 24-hour number:

- a temperature above 37.5°C
- a temperature below 36°C
- you feel unwell, even with a normal temperature
- you have symptoms of an infection.

Symptoms of an infection include:

- feeling shivery and shaking
- a sore throat
- a cough
- breathlessness
- diarrhoea
- needing to pass urine (pee) often, or discomfort when you pass urine.

**It is important to follow any specific advice your cancer treatment team gives you.**

Your white blood cell count will usually return to normal before your next treatment. You will have a blood test before having more treatment. If your white blood cell count is low, your doctor may delay your treatment for a short time, until your cell count increases.

You may be given a drug called [G-CSF](#). This encourages the body to make more white blood cells. You have it as a small injection under the skin.

## Bruising and bleeding

This treatment can reduce the number of platelets in your blood. Platelets are cells that help the blood to clot.

If the number of platelets is low, you may bruise or bleed easily. You may have:

- nosebleeds
- bleeding gums
- heavy periods
- blood in your urine (pee) or stools (poo)
- tiny red, brown or purple spots that may look like a rash – these spots can be harder to see if you have black or brown skin.

If you have any unexplained bruising or bleeding, contact the hospital straight away on the 24-hour number. You may need a drip to give you extra platelets. This is called a platelet transfusion.

## Anaemia (low number of red blood cells)

This treatment can reduce the number of red blood cells in your blood. Red blood cells carry oxygen around the body. If the number of red blood cells is low, this is called anaemia. You may feel:

- very low in energy
- breathless

- dizzy and light-headed.

If you have these symptoms, contact the hospital straight away on the 24-hour number. You may need treatment for anaemia. If you are very anaemic, you may need a drip to give you extra red blood cells. This is called a blood transfusion.

## Numb or tingling hands or feet (peripheral neuropathy)

This treatment may affect the nerves. This can cause numb, tingling or painful hands or feet. You may find it hard to fasten buttons or do other fiddly tasks.

Tell your doctor straight away if you have these symptoms. Your doctor may need to lower the dose of the drug. The symptoms usually improve slowly after treatment finishes. But for some people they may never go away. Talk to your doctor if you are worried about this.

## Effects on the lungs

This treatment can cause changes to the lungs. Tell your doctor if you develop:

- a cough
- wheezing
- a fever (high temperature)
- breathlessness, even when doing very little.

You should also tell them if any existing breathing problems get worse. You may have tests to check your lungs.

If your symptoms start suddenly, contact the hospital straight away. You may have inflammation of the lungs. This is called pneumonitis. Your doctor can tell you more about this.

## Feeling sick

Your doctor, nurse or pharmacist will prescribe anti-sickness drugs to help prevent or control sickness. Take the drugs exactly as they tell you to, even if you do not feel sick. It is easier to prevent sickness than to treat it after it has started.

[If you feel sick](#), take small sips of fluid often and eat small amounts regularly. It is important to drink enough fluids. If you continue to feel sick, or if you are sick (vomit) 1 to 2 times in 24 hours, contact the hospital on the 24-hour number as soon as possible. They will give you advice. They may change your anti-sickness treatment. Let them know if you still feel sick.

## Diarrhoea

This treatment may cause [diarrhoea](#). Diarrhoea means passing more stools (poo) than is normal for you, or having watery or loose stools. You may also have stomach cramps. If you have a stoma, it may be more active than usual.

If you are passing loose stools 3 or more times a day and this is not normal for you, contact the hospital as soon as possible on the 24-hour number. Follow the advice they give you about:

- taking anti-diarrhoea medicines
- drinking enough fluids to keep you hydrated and to replace lost salts and minerals

- any changes to your diet that might help.

They might also ask you for a specimen of your stool to check for infection.

## Constipation

This treatment can cause [constipation](#). Constipation means that you are not able to pass stools (poo) as often as you normally do. It can become difficult or painful. Here are some tips that may help:

- Drink at least 2 litres (3½ pints) of fluids each day.
- [Eat high-fibre foods](#), such as fruit, vegetables and wholemeal bread.
- Do regular gentle exercise, like going for short walks.

If you have constipation, contact the hospital on the 24-hour number for advice. They can give you drugs called laxatives to help.

If you have not been able to pass stools for over 2 days and are being sick, contact the 24-hour number straight away.

## Feeling tired

[Feeling tired](#) is a common side effect of this treatment. It is often worse towards the end of treatment and for some weeks after it ends. Try to pace yourself and plan your day so you have time to rest. Gentle exercise, like short walks, can help you feel less tired.

If you feel sleepy, do not drive or use machinery.

## Hair loss

Your hair will get thinner. Or you may lose all the hair from your head. You may also lose your eyelashes and eyebrows, as well as other body hair. Hair loss usually starts after your first or second treatment.

If you want to cover up hair loss, there are different ways you can do this. Your nurse will give you information about [coping with hair loss](#).

Remember to [protect your skin from the sun](#). Use suncream with a sun protection factor (SPF) of at least 30 on your scalp. Or cover up with a hat or scarf.

Hair loss is almost always temporary. Your hair will usually grow back after treatment ends.

## Sore mouth and throat

This treatment may cause a [sore mouth and throat](#). You may also get mouth ulcers. This can make you more likely to get a mouth or throat infection. Use a soft toothbrush to clean your teeth or dentures in the morning, at night and after meals.

Contact the hospital straight away on the 24-hour number, if:

- a sore mouth or throat affects how much you can drink or eat
- your mouth, tongue, throat or lips have any blisters, ulcers or white patches.

They can give you advice, and mouthwash or medicines to help with the pain or to treat any infection. Follow their advice and make sure you:

- drink plenty of fluids
- avoid alcohol and tobacco
- avoid food or drinks that irritate your mouth and throat.

## Changes to your taste

Some foods may taste different or have no taste. Try different foods to find out what tastes best to you. You may also get a bitter or metallic taste in your mouth. Your doctor, nurse or pharmacist can give you advice. It might help to try:

- sucking sugar-free sour or boiled sweets
- eating cold foods
- eating sharp-tasting fresh fruit.

Taste changes usually get better after treatment ends. We have more information about [coping with changes to taste](#).

## Loss of appetite

This treatment can affect your appetite. Don't worry if you do not eat much for 1 or 2 days. But if your appetite does not come back after a few days, or if you are losing weight, tell your doctor, nurse or pharmacist. They can give you advice. They may give you food or drink supplements. Or they may suggest changes to your diet or eating habits to help.

## Tummy pain or indigestion

[Steroids](#) can irritate the stomach lining. Tell your doctor, nurse or pharmacist if you have pain in your tummy or indigestion. They can give you drugs to help. Taking steroids with food can also help to protect your stomach.

## Skin changes

This treatment may [affect your skin](#). It may cause a rash, which might be itchy. Your doctor or nurse can tell you what to expect. If your skin feels dry, try using an unperfumed moisturising cream every day.

During treatment, and for several months afterwards, you will be more sensitive to the sun. Your skin may burn more easily than usual. You can still go out in the sun, but use a suncream with a sun protection factor (SPF) of at least 30. Cover up with clothing and a hat.

Your skin may darken. It will return to its normal colour after you finish treatment. If you have had radiotherapy (either recently or in the past), the area that was treated may become red or sore.

Very rarely, this treatment may cause a more serious skin condition. You may have a skin rash which then blisters, and your skin can peel. You may also feel unwell with flu-like symptoms such as a high temperature and joint pain. If you have any of these symptoms, call your doctor or the 24-hour number the hospital has given you.

Always tell your doctor or nurse about any changes to your skin. They can give you advice and may prescribe creams or medicines to help. Skin changes usually improve after treatment finishes.

## **Raised blood sugar levels**

Steroids may raise your blood sugar levels. Symptoms of raised blood sugar include:

- feeling thirsty
- needing to pass urine (pee) more often than usual
- often feeling tired.

Tell your doctor or nurse if you have these symptoms. If your blood sugar is high, your doctors may give you drugs to lower it. Your blood sugar usually goes back to normal after you have finished your treatment.

If you have [diabetes](#), your blood sugar levels may be higher than usual. You may need to check them more often, or make changes to your diabetic treatment. Your diabetic doctor or nurse will talk to you about how to manage this.

## **Muscle or joint pain**

You may get pain in your muscles or joints for a few days after treatment. If this happens, tell your doctor, nurse or pharmacist. They can give you painkillers and advice. They can also tell you if any of the painkillers you usually take are suitable.

Tell them if the pain does not get better. Having warm baths and resting regularly may help.

## **Bladder irritation**

Cyclophosphamide may irritate your bladder and cause discomfort when you pee (pass urine).

- Drink at least 2 litres (3½ pints) of fluids every day, especially during the 24 hours after chemotherapy.
- Empty your bladder regularly and to try to pee as soon as you feel the need to.
- Contact the hospital straight away if you feel any discomfort or stinging when you pee, or if you notice any blood in your urine.

## **Fluid build-up**

Steroids can cause a build-up of fluid in the body. This is more common if you are taking them for a long time. This will slowly get better after treatment ends. Contact the hospital on the 24-hour number if you:

- are gaining weight
- have swelling in your face, legs or ankles.

They can give you and advice and treatment to help.

## Pink or red urine

Your urine may be pink or red for up to 48 hours after you have treatment. This is usually due to the colour of doxorubicin. It is not harmful. But always check with your doctor, nurse or pharmacist if you have any concerns.

## Less common side effects

### Effects on the heart

This treatment can affect how the heart works. You may have [tests](#) to check how well your heart is working. These may be done before, during and after treatment.

If the treatment is causing [heart problems](#), your doctor may change the type of treatment you are having.

Contact the hospital straight away on the 24-hour number if you have any of these symptoms during or after treatment:

- breathlessness
- dizziness
- changes to your heartbeat (palpitations)
- swollen feet and ankles.

Other conditions can cause these symptoms, but it is important to get them checked by a doctor.

Always call **999** if you have:

- chest pain, pressure, heaviness, tightness or squeezing across the chest
- difficulty breathing.

### Effects on the nervous system

Rarely, this treatment can affect the nervous system. It can cause a condition called Progressive multifocal leukoencephalopathy (PML). Symptoms include:

- difficulty speaking
- difficulty walking
- memory loss
- confusion
- changes in your eyesight.

Contact the hospital straight away on the 24-hour number if you have any of these symptoms or if someone with you notices you have these symptoms.

It is important not to drive or operate machinery if you have these symptoms.

## Pancreatitis

Rarely, brentuximab may cause inflammation of the pancreas. If you have pain in your tummy (abdomen), tell your doctor straight away.

## Mood and behaviour changes

This treatment can affect your mood. You may:

- feel anxious or restless
- have mood swings
- have problems sleeping.
- Taking your steroids in the morning may help you sleep better.

If you have any of these side effects, tell your doctor, nurse or pharmacist. They may make some changes to your treatment if the side effects become a problem.

## Effects on the liver

This treatment can change the way your liver works. This usually goes back to normal after treatment finishes. Rarely, the drug may cause your skin and the whites of your eyes to turn yellow. This is called jaundice. You will have regular blood tests to check how well your liver is working.

## Tumour lysis syndrome (TLS)

Some people are at risk of developing a condition called tumour lysis syndrome (TLS). The risk is highest at the beginning of treatment. TLS happens when treatment makes large numbers of cancer cells die and break down quickly. This releases lots of waste products into the blood and can affect the kidneys and heart.

TLS can be prevented or treated. You will have regular blood tests to check for TLS. You may have:

- extra fluids through a drip
- medicines such as allopurinol tablets or [rasburicase](#) through a drip.

Drinking at least 2 litres (3½ pints) of fluid a day will also help.

## Second cancer

This treatment can increase the risk of developing a second cancer years later. This is rare. The benefits of treatment usually far outweigh this risk. Your doctor, nurse or pharmacist can give you more information.

## Other information

### Blood clot risk

Cancer and some cancer treatments can increase the risk of a [blood clot](#). Contact the hospital straight away on the 24-hour number if you have any of these symptoms during or after treatment:

- throbbing pain or swelling in a leg or arm
- reddening of the skin in the area – if you have black or brown skin, this can be harder to notice, but the skin might become darker
- suddenly feeling breathless or coughing.

Always call **999** if you have:

- chest pain
- difficulty breathing.

A blood clot is serious, but it can be treated with drugs called anticoagulants. These thin the blood. Your doctor, nurse or pharmacist can give you more information about preventing and treating blood clots.

## Vaccinations

Cancer doctors usually recommend that people with cancer have vaccinations for flu and [coronavirus \(covid\)](#). They may also recommend other vaccines, such as Shingrix<sup>®</sup> for shingles. These all help reduce your risk of serious illness from these infections. Most people can have these vaccines, including people with weak immune systems.

You should not have live vaccines if your immune system is weak. This includes if you are having or recently had chemotherapy, radiotherapy or other cancer treatments that affect your immune system. Live vaccines can make you unwell because they contain a very weak version of the illness they protect you against. There are several live vaccines, including the yellow fever vaccine.

It is important to ask your doctor, nurse or pharmacist for advice about having vaccinations. They can explain what vaccines are right for you and when it is best to have them.

## Other medicines

Some medicines can affect how this treatment works or be harmful while you are having it. Always tell your cancer doctor, nurse or pharmacist about any drugs you are taking or planning to take, such as:

- medicines you have been prescribed
- medicines you buy in a shop, pharmacy or online
- vitamins or supplements
- [herbal drugs and complementary or homeopathic therapies](#)
- recreational drugs – for example, cannabis.

## Grapefruit and grapefruit juice

During treatment with cyclophosphamide, you should avoid grapefruit and grapefruit juice. This is because they can increase the side effects of cyclophosphamide.

## Contraception

Your doctor, nurse or pharmacist will advise you not to [get pregnant](#) or [make someone pregnant](#) while having this treatment and for some time afterwards. The drugs may harm a developing baby. It is important to use contraception to prevent pregnancy. Follow their advice about:

- what types of contraception to use
- how long after treatment you should continue to use contraception.

## Breastfeeding

You are advised not to [breastfeed](#) while having this treatment, or for some time after treatment ends. This is because the drugs could be passed to the baby through breast milk.

Your doctor, nurse or pharmacist can give you more information.

## Fertility

Some cancer drugs can affect whether you can [get pregnant](#) or [make someone pregnant](#). If you are worried about this, it is important to talk with your doctor before you start treatment.

## Changes to periods

If you have periods, these may become irregular or stop while you are having this treatment. They might return after treatment, but this does not always happen. Your [menopause](#) may start sooner than it would have done. Your doctor, nurse or pharmacist can give you more information.

## Sex

It is possible that small amounts of chemotherapy may be passed on through vaginal fluids or semen. If you have sex in the first few days after treatment, your cancer team will usually advise using condoms or a dental dam to protect your partner.

## Medical and dental treatment

If you need medical treatment for any reason other than cancer, always tell the healthcare professional that you are having cancer treatment. Give them the contact details for your cancer doctor or cancer team so they can ask for advice.

If you have appointments with a dentist, always tell them you are having cancer treatment. Talk to your cancer team before you have any dental treatment.

## About our information

This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer.

## References



Visit the [electronic Medicines Compendium \(eMC\)](#) to download a Patient Information Leaflet (PIL) for more detailed information. The leaflet lists all known side effects.

[Cancer Information Development team](#)

Author

## Date reviewed

Reviewed: 01 January 2024 | Next review: 01 January 2026

ⓘ This content is currently being reviewed. New information will be coming soon.

Our cancer information meets the [PIF TICK](#) quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about [how we produce our information](#).

## The language we use

We want everyone affected by cancer to feel our information is written for them.

We want our information to be as clear as possible. To do this, we try to:

- use plain English
- explain medical words
- use short sentences
- use illustrations to explain text
- structure the information clearly
- make sure important points are clear.

We use gender-inclusive language and talk to our readers as 'you' so that everyone feels included. Where clinically necessary we use the terms 'men' and 'women' or 'male' and 'female'. For example, we do so when talking about parts of the body or mentioning statistics or research about who is affected.

You can read more about how we produce our information [here](#).

## How we can help

### How to print a PDF from this website

Instructions on how you can make a PDF version of this page using the print function on your browser.

### Macmillan Support Line

The Macmillan Support Line is a free and confidential service for people living with and affected by cancer. If you need to talk, we'll listen.



### **Call us free on 0808 808 0000**

8am to 8pm, 7 days a week. You can talk to our cancer specialists about anything.



### **Email the Macmillan Support Line**

Email the Macmillan Support Team if you have any questions.

### **Chat online**

Available 8am to 7.45pm, 7 days a week. For confidential, real-time support if you don't want to call.



### **Online Community**

Chat online anonymously to others who understand what you are going through. Our community is available 24/7 and has dedicated forums where you can get advice and ask our experts.



### **Find cancer support services in your area**

What's going on near you? Find out about support groups, where to get information and how to get involved with Macmillan where you live.

**Get help**